

24 Feb 2024

Subject: Synexis LLC Comments in Response to the Request for Information Regarding Test Procedure and Energy Conservation Standards for Consumer Air Cleaners *Docket Numbers EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036*

Dear Sir or Madam,

Synexis LLC respectfully submits this letter in response to the U.S. Department of Energy's (DoE) tentative determination that air cleaners qualify as a covered product, published in the *Federal Register* on September 16, 2021.

Synexis LLC is a small business developing, manufacturing, and commercializing indoor air cleaning devices in the United States. Synexis strives to provide its customers safe and effective devices that help to improve indoor air quality by providing an added layer of protection when coupled with standard cleaning and disinfection practices. Because Synexis devices meet the inclusion criteria established in the NOPD September 2021 definition of air cleaners, we have a vested interest in the outcome of the proposed ruling and offer the following general comments to identified sections of the RFI.

Definitions

- *DoE must revise and further clarify its definition of consumer air cleaners*

The proposed DOE definition for consumer air cleaner/s as it currently stands does not account for all of the various technologies that exist in this space and could benefit from further clarification. The points of clarification and differentiation described below become critical to the discussion as DoE considers appropriateness of various test methods and standards to be included in the scope of the proposed ruling.

While we agree with points (1) and (2) in the proposed definition and also agree with points (3) and (4) in that a consumer air cleaner removes, destroys, or deactivates particulates and microorganisms from the air by means other than UV light; points (3) & (4) could be further clarified by adding information regarding mechanism/s of action and claims associated with these devices. For example:

- Devices that incorporate HEPA, Charcoal, carbon, MERV or other filters of a pore size sufficient to remove and/or destroy particulates and microorganisms from the air and/or utilize Photocatalytic Oxydation, Bipolar ionization, Dry Hydrogen Peroxide, or other similar technologies with or without additional filtration as a means to remove, destroy or deactivate particulates and microorganisms from the air

-Devices intended to operate in the presence of human (or other living) occupants on a continuous basis.

While devices claiming UV light as their mode of action are a reasonable exclusion, these specific examples of device types that would be **in scope** would provide additional clarity to this section of the definition.

With regard to Definition point (5), it is our opinion that portable „air conditioners“ or similar devices that incorporate HEPA filtration with any supplemental claims related to cleaning the air (in addition to

cooling) should not be excluded from the definition of consumer air cleaners. Addition of HEPA, MERV or other filtration mechanisms to portable air conditioners that carry air cleaning claims and meet other defining characteristics should not be excluded from the scope of the ruling.

Test Procedure for Consumer Air Cleaners

- *DOE requests comments on whether ANSI/AHAM AC-1-2020 provides an appropriate method to use as the basis for a Federal test method and for defining energy conservation standard levels for consumer air cleaners.*

Response: The ANSI/AHAM method referenced above is not currently a statutory requirement for indoor air cleaning devices and was designed for the purpose of measuring performance of indoor air cleaners which focus on removal of particulate matter from the air, presumably via filtration mechanisms. When taking into account the proposed definition of indoor air cleaners, this method does not account for performance measurements of devices that use mechanisms of action other than filtration. If the proposed revised definition of indoor air cleaners stands, this test method alone is not appropriate due to the various mechanisms and technologies that exist within the air cleaning device space.

- *DOE requests comment on the use of the CADR, as opposed to another metric such as rate of decay, to characterize consumer air cleaner performance. In particular, DOE requests comment on whether consumers could find the unit of measurement of cfm for CADR confusing and misunderstand it as referring to the rate of air movement through the device.*

Response: If the proposed definition of indoor air cleaners stands, solely using CADR as the defining performance metric does not account for devices that operate by methods other than filtration. It is our opinion that an appropriate metric would take into account the mechanism of action of the air cleaner as well as the various types of contaminants the devices claim to remedy (e.g. particulate, microorganisms, mold, odors, volatile organic compounds (VOCs), etc).

- *DOE requests comment on whether the power measurement could vary based on the particulate test that is used to measure operating power. If power measurement varies based on the particulate test, DOE requests comment on which particulate test (pollen, dust, or cigarette smoke) should be used as the basis for the power measurement in any Federal test procedure that DOE may develop. Alternately, DOE requests comment on whether it should consider requiring power measurements for each particulate test and use a simple or weighted average to determine operating power.*

Response: Measuring energy consumption during individual particle mitigation testing would not provide data representative of real world scenarios. Device energy consumption may vary based on size of particles used in particulate tests. Particulates of various sizes can cause filters to become entrained with pollutant particles and require greater pressure to move air through the device. Additionally, devices that do not utilize filtration as the primary means of cleaning the air risk failure of the performance testing that is aimed at demonstrating filtration effectiveness. It is our opinion that DOE should consider performance testing that considers not only particle testing but testing that incorporates appropriate metrics to account for the mechanism of action of the air cleaner as well as performance against the various types of contaminants the devices claim to remedy (e.g. particulate matter, microorganisms, mold, odors, volatile organic compounds (VOCs), etc). Finally, we do not agree that power measurements for each particulate test would be representative of real world energy consumption nor provide any useful data.

- *DOE requests comment on whether it should consider testing consumer air cleaners at any other power level in addition to the maximum power level required by ANSI/AHAM AC-1-2020.*

Response: It is our opinion that testing at maximum power levels is appropriate as it represents a worst-case scenario with regard to energy consumption. Medium and low power settings are likely to exhibit different performance characteristics in different devices and would not provide an appropriate metric to compare different air cleaners.

- *DOE requests comment on whether ANSI/AHAM AC-1-2020 could also be used to test other types of consumer air cleaners, such as ceiling-mounted products*

Response: The location of the air cleaning device is less critical than the mechanism of action of the device when testing performance and/or energy consumption. Whether the devices are free-standing, portable devices, wall-mounted, or ceiling mounted, the power utilization and performance should remain the same. It is our opinion that any test method applied to indoor air cleaning devices must be appropriate for portable as well as mounted or fixed devices.

- *DOE requests comment on the suitability of the standby power measurement procedure specified in ANSI/AHAM AC-1-2020, IEC 62301 Ed. 2.0, or any other test method for measuring standby mode and off mode energy use of consumer air cleaners, in light of EPCA's requirement in 42 U.S.C. 6295(gg)(2)(A)) for DOE to consider the most current version of IEC Standard 62301.*

Response: There are a number of devices in the air cleaning market that are intended to operate continuously. It is our opinion that while a measurement of standby power utilization may provide some baseline data with regard to energy consumption, maximum energy utilization will occur while the devices are in operation.

Other test methods for consideration

- *DOE requests comment on whether it should consider any methodology for measuring the removal efficacy of microorganisms (i.e., viruses, bacteria, mold, etc.) from indoor air as part of a Federal test procedure for consumer air cleaners.*

Response: Provided the modified definition of air cleaning devices stands, it is critical for DoE to consider performance measurement methods that account for all of the various technologies available. Of the test methods proposed, no single method accounts for all of the technologies included in the proposed definition of air cleaners. It is our opinion that multiple test methods should be employed. For example, CADR and ANSI-AHAM AC-1 test only measure particulate reduction in a 20 minute time period. Devices with high cfm output and lack of carbon and fine particle filters tend to perform better. Higher end purifiers with carbon and HEPA/Merv filters typically run at lower cfms but receive lower CADR rating at the 20 minute mark. Particulate reduction is a reasonable parameter to consider however, it does not cover all of the device types and should not be the sole testing criteria.

AHAM recently published a white paper and new guidance AHAM-AC-5-2022 (2022) describing a proposed methodology for assessing the reduction rate of key bioaerosols by portable air cleaners using an aerobiology test chamber. This proposal may warrant further consideration by DoE as an interim measure until this and other methods are developed, identified and evaluated. Again, appropriateness of this method should be evaluated dependent upon the final definition of air cleaning devices and prior to implementation should be determined to be appropriate for each type of device falling under the definition.

- *DOE requests comment on the suitability of each of the RTI Test Method and the KTL Test Method for measuring a consumer air cleaner's removal efficacy of microorganisms from indoor air.*

Response: KTL method only tests for bacterial reduction. This test does not account for devices that claim to mitigate other types of contaminants.

The RTI test method if used in combination with some additional test methods (NRCC or others) would provide better evidence of the effectiveness of devices claiming effectiveness against various types of contaminants as it includes bacteria, virus and mold reduction.

- *DOE requests comment on the additional test methods identified in this section that measure the performance of consumer air cleaners using various particulates. In particular, DOE requests comment on the scope, methodology, and types of particulates, pollutants, and/or microorganisms that are included in each test method.*

Response: Given the recent worldwide pandemic, there has been increased interest in filtration type devices that claim to mitigate viruses, bacteria, mold and VOC's. However there has been a lot of misinformation circulating and lack of uniform performance, safety, and effectiveness testing parameters has lead to significant consumer confusion in this space.

For these reasons, and in consideration of the proposed definition of air cleaners, we feel it is important to include elements of several of the existing testing methods mentioned above. For example, the Research Triangle Institute (RTI) and National Research Council Canada (NRCC) capture many of the effectiveness criteria as RTI measures airborne virus, bacteria and mold reduction while NRCC method measures VOC and Ozone reduction. From a safety perspective, the NRCC method would demonstrate that the devices are not producing harmful level of by-products. Additionally, UL or other standardized electrical safety test methods should be considered.

- *DOE requests comments on whether any other test methods have been developed for consumer air cleaners that would be relevant to DOE's consideration of a Federal test procedure to measure the energy efficiency of consumer air cleaners. In particular, DOE seeks comment on test methods that could be used to test "non-portable" consumer air cleaners, such as those that are permanently mounted (e.g., ceiling-mounted air cleaners) or that provide whole-home air cleaning in conjunction with central heating or air conditioning systems; and test methods that could be used to measure the performance of consumer air cleaners that destroy or deactivate contaminants from the air instead of removing them.*

Response: Here again, any test methods considered should be around energy consumption, effectiveness, and safety and the physical location of the devices. The same methods established for the portable devices should be applied to fixed or mounted devices.

- *DOE requests comment on the technical feasibility of integrating measures of standby mode and off mode energy consumption into the overall energy efficiency descriptor (i.e., creating an integrated metric) for consumer air cleaners. In particular, DOE requests comment on its example approach of defining an integrated CADR/W metric, in which the denominator would represent a weighted average of the power consumption associated with active mode, standby mode, and off mode, weighted by the amount of time spent in each mode.*

Response: CADR test chambers are not representative of actual room sizes, and therefore CADR ratings do not reflect actual usage under real-world conditions.

At 12ft x 12ft x 7ft (1,008cuft), the CADR test chamber is the equivalent volume of a 10ft x 10ft room with 10ft ceiling (1,000cuft). Air purifiers are generally used in common areas, living rooms, waiting rooms, conference rooms, etc. These areas range from 500sf to 1000sf with 10ft ceiling, or 5,000cuft to 10,000cuft.

Testing in large chambers (5,000cuft to 10,000cuft) is more appropriate as a chamber of this size is more representative of real world conditions. Additionally, equipment efficiency should be rated on SqFt/W.

CADR only accounts for particle removal presumably via filtration. As there are other modalities for air cleaning, and considering the proposed definition, there must be a method employed that also tests effectiveness against virus, bacteria, mold, VOCs and other by-products.

A device with no carbon or HEPA filter designed to treat a 100sf room and consumes a 35W might have a better CADR rating than a device with carbon and HEPA filters designed to treat a 1,000sqft room and consumes 200w. Under this condition the first device would have a lower CADR/W ratio. However, the first device would consume almost twice the power per SqFt of the second and not mitigate VOCs, virus, bacteria and others. For these reasons, using CADR/W alone is not an appropriate metric as it might be misleading with regard to overall energy consumption required to achieve maximum effectiveness.

- *DOE requests comment on consumer usage of consumer air cleaners, in particular, the amount of time spent in active mode, standby mode, and off mode.*

Response: There are a number of devices in the air cleaning market that are intended to operate continuously. It is our opinion that while a measurement of standby power utilization may provide some baseline data with regard to energy consumption, maximum energy utilization will occur while the devices are in operation.

On the other hand, if standby modes were to be tested, a 20 minute CAdR test would not provide an ample test period. Test time period would probably need to increase to 24 hours minimum to provide an adequate baseline.

- *DOE requests comment on whether cigarette smoke would be the appropriate particulate for determining a CADR rating of air cleaners under a DOE test procedure, should DOE adopt a measurement of CADR in a test procedure for consumer air cleaners. If cigarette smoke is not the most appropriate particulate, DOE requests comment on other particulate(s) that would be more appropriate as the basis for measurement, including data and information to support such a recommendation.*

Response: If the proposed definition for air cleaners stands and includes not only devices that remove particulates but also devices that destroy, remove or deactivate microorganisms (e.g. bacteria, viruses, molds, etc), DoE should consider measurement methods that account for various mechanisms of action utilized to achieve this end. Removal of particulate matter as a sole means for measurement of effectiveness of air cleaning devices is not sufficient to cover all technologies meeting the proposed definition.

- *DOE requests comment on whether the CADR/W efficiency metric would be appropriate for characterizing the energy efficiency of consumer air cleaner units permanently mounted to a structure.*

Response: As stated previously, the location of the air cleaning device is less critical than the mechanism of action of the device when testing performance and/or energy consumption. Whether the devices are free-standing, portable devices, wall-mounted, or ceiling mounted, the power utilization and

performance should remain the same. It is our opinion that any test method applied to indoor air cleaning devices must be appropriate for portable as well as wall-mounted or fixed devices. We feel a SqFt/W would be a more representative metric.

- *DOE requests comment on whether the CADR/W metric would be appropriate for consumer air cleaners that clean the air by destroying or deactivating particulates and microorganisms from the air instead of removing them.*

Response: As not all technologies utilize filtration methodologies as their mode of action, CADR is unfortunately not a metric that covers all air cleaning devices/technologies and their various modes of action. Our opinion is that if a standard is established by DoE, it should incorporate a metric that is appropriate across all air cleaning technologies exclusive of their mode of action. We feel a SqFt/W would be a more representative metric.

- *DOE requests comment on whether any other metrics not already discussed in this RFI would provide a better measure of energy efficiency or energy use of consumer air cleaners during a representative average use cycle or period of use.*

Response: Our opinion is that air cleaning technologies are only one facet of a total systems approach to indoor air quality. A more comprehensive approach would include an assessment of the overall energy consumption required to obtain optimal indoor air quality. The recommended approach from ASHRAE/ANSI as well as agencies such as CDC include various interventions including, ventilation, HVAC system maintenance, air cleaning devices, HEPA and other modalities. To properly assess the energy consumption associated with the air cleaning devices, it becomes necessary to consider not only the air cleaning devices themselves but all other devices in the environment that are working to improve the indoor air quality. There are air cleaners that reduce the burden on HVAC and other technologies and by way of their introduction in the environment have the potential to reduce the energy consumption overall. For example:

- The use of appropriate self-contained filters can reduce overall building power consumption.
- Current pandemic guidelines suggest fresh air intakes be increased to the maximum settings, the use of MERV 13 or higher filters, exhaust fans be used at their maximum levels, and that HVAC systems run 2 hours before and after buildings are occupied.
- These measures significantly increase operational and energy costs by:
 - o Reducing equipment lifespan
 - o Increasing energy usage required for filtration
 - o Increasing energy to heat/cool additional fresh air (up to 5 times; from 5% to 25% fresh air)
 - o Additional operating hours to flush out air.
- Mitigating viruses, bacteria, mold, VOC's and other contaminants through the use of self-contained units can reduce outside air requirements to minimum levels, eliminate MERV 13 filters and provide other energy and operational savings which can offset filter's installation and operational costs.

Energy Conservation Standard

- *DOE seeks comment on whether energy conservation standards for consumer air cleaners would be economically justified, technologically feasible, and would result in a significant savings of energy.*

Response:

Economically justified: There is currently a lot of misinformation about product use and performance with very little government oversight. Some organizations operating in this space are investing

considerable time and resources to test and validate effectiveness, efficiency, and safety of their products. Implementation of uniform methods and standards would likely reduce costs by standardizing evaluation processes and will also provide common criteria on which consumers can make informed decisions around which technologies they wish to utilize.

Technologically feasible: There are a number of test methodologies and evaluation criteria that could be modified to meet the testing needs for development of a standard. The most important consideration is that any testing methodologies are appropriate to test any air cleaners that meet the proposed definition.

Energy savings: As air cleaning devices become more common, it is important for consumers to understand their power implications; a 35W device that covers 100sqft might be significantly more expensive to operate than a 200W devices that covers 1,000sqft. Again, a systemic approach to evaluating total energy consumption required to achieve and maintain optimal indoor air quality would be most appropriate.

Product Classes

- *DOE requests comment on whether capacity or any other performance-related features, such as air cleaning technology (i.e., whether the product destroys or deactivates contaminants from the air or removes them), of consumer air cleaners would justify the establishment of different product classes (i.e., would justify different standards for such classes).*

Response: Definition of a standard should be applicable to all devices operating in the air cleaning technology space. Sub-classes would likely confuse the issue and be difficult to apply equally across all technologies.

Technology Assessment

- *For each technology option suggested by stakeholders, DOE seeks information regarding its market adoption, costs, and any concerns with incorporating the technology into products (e.g., impacts on consumer utility, potential safety concerns, manufacturing or production challenges, etc.).*

Response: There are air cleaning devices that purposely generate ozone or other substances as their mechanism of action. As Ozone above established thresholds is known to be harmful to humans, there should be safety standards considered for determination of hazardous by-products across all air cleaning technologies.

Screening Analysis

- *DOE requests feedback on whether any air cleaner technology options would be screened out based on the five screening criteria described in this section. DOE also requests information on the technologies that would be screened out and the screening criteria that would be applicable to each screened out technology option.*

Response: It is our opinion that this criterion should be eliminated since it is in direct and fundamental conflict with Sole Source rights established through US laws to protect intellectual property rights granted by the US Patent and Trademark office. If the US Government grants monopolistic rights to certain technology options through the patent process, it seems strange that the DoE would independently eliminate those same technology options.

Thank you for your consideration of the comments above.



Respectfully submitted on behalf of Synexis, LLC,

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